Type MS · Series made of logarithms

Answer 1 of 3 · Question 4 part whole · 5 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part whole in this mark scheme extract

number		
4	$u_1 = (1+1)\ln 4 = 2\ln 4$ and $d = \ln 4$	M1
	$S_n = \frac{n}{2}\left(2\times '2\ln 4'+(n-1)'\ln 4'\right)$ or $S_n = \frac{n}{2}\left('2\ln 4'+(n+1)'\ln 4'\right)$	M1
	$\ln 4$ to either $2\ln 2$ or $\ln 2^2$ at any stage.	M1
	$S_n = \frac{n}{2}(2n+6)\ln 2$ or $S_n = \frac{n}{2}(n+3)\ln 4$ or $S_n = \frac{n}{2}\left(\ln 2^6 + \ln 2^{2n}\right)$ or $S_n = \frac{n}{2}\left(\ln 4^3 + \ln 4^n\right)$	M1
	$S_n = \ln 2^{n^2+3n}$	A1 cso
		Total 5 marks

Type MS · Series made of logarithms

Answer 1 of 3 · Question 4 part whole · 5 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Notes
M1	Finds the first term and the common difference. $u_1 = (1+1)\ln 4 = 2\ln 4$ and $d = \ln 4$ Both must be correct for this mark.
	The general principle of marking this question is as follows: **Note: Their *a* and *d* must be in terms of ln 4 or 2ln 2** • Second M1 for a correct substitution of their a and their d into either $\frac{n}{2}(2a+(n-1)d)$ or $\frac{n}{2}(a+L)$ • Third M1 is for dealing **correctly** with all terms in ln 4 at any stage $(\ln 4 = 2\ln 2$ or $\ln 2^2)$ seen anywhere in the solution. • Fourth M1 for attempting to simplify the sum to the required form using their a and d • Final mark is for obtaining the given answer with no errors seen.
M1	Uses either form of the summation formula for an arithmetic series with their a and d provided both are in terms of ln 4 or 2ln 2 **There must be no errors in the use of and substitution of their values into the formula for this question** – it is given on page 2. $S_n = \frac{n}{2}(2\times'2\ln 4'+(n-1)'\ln 4')$ or $S_n = \frac{n}{2}('2\ln 4'+(n+1)'\ln 4')$
M1	For **correctly** changing all terms in ln 4 to either 2ln 2 or $\ln 2^2$ at any stage. You may see this step at the end of the solution.
M1	Simplifies their expression in either **ln 2 or ln 4** to obtain one of the following. $S_n = \frac{n}{2}(2n+6)\ln 2$ or $S_n = \frac{n}{2}(n+3)\ln 4$ or $S_n = \frac{n}{2}(\ln 2^6 + \ln 2^{2n})$ or $S_n = \frac{n}{2}(\ln 4^3 + \ln 4^n)$
A1cso*	For obtaining the given answer in full with no errors. $S_n = \ln 2^{n^2+3n}$

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Answer 1 of 3 · Question 4 part whole · 5 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Series made of logarithms

Answer 2 of 3 · Question 3 part whole · 7 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

Number		
3	$\ln 12 = \ln a + (2-1)\ln b$ oe	M1
	$ab = 12$ oe	A1
	$\ln 768 = \ln a + (5-1)\ln b$ oe	M1
	$ab^4 = 768$ oe	A1
	$\frac{768}{12} = \frac{ab^4}{ab}$ $(b^3 = 64)$	ddM1
	$b = 4 \quad a = 3$	A1 A1
	ALT 1	
	$\ln a + (2-1)\ln b = \ln 12$ oe	M1 A1
	$\ln a + (5-1)\ln b = \ln 768$ oe	
	$3\ln b = \ln b^3$ $\ln 768 - \ln 12 = \ln 64$	M1 A1
	$b^3 = 64$	ddM1
	$b = 4 \quad a = 3$	A1 A1
	ALT 2	
	$d = \ln 12 - \ln a$	M1
	$(d = \ln b = \ln 12 - \ln a \Rightarrow \ln b = \ln\left(\frac{12}{a}\right)) \Rightarrow b = \frac{12}{a}$	A1
	$\ln 768 = \ln a + \ln\left(\frac{12}{a}\right)^4$	M1
	$\ln 768 = \ln\left(\frac{12^4}{a^3}\right)$	A1
	$a^3 = \frac{20736}{768}$	ddM1
	$b = 4 \quad a = 3$	A1 A1
	ALT 3	
	$(u_2 =) \quad u_1 + d = \ln 12$	M1 A1
	$(u_5 =) \quad u_1 + 4d = \ln 768$	
	$3d = \ln 768 - \ln 12$	M1
	$d = \ln 4$	A1
	$u_1 = \ln 12 - \ln 4 = \ln 3 \ (= \ln a)$	ddM1
	$b = 4 \quad a = 3$	A1 A1
		[7]

Type MS · Series made of logarithms

Answer 2 of 3 · Question 3 part whole · 7 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

Part	Mark	Additional Guidance
	M1	Correct equation as shown oe
	A1	Correct equation as shown oe
	M1	Correct equation as shown oe
	A1	Correct equation as shown oe
	ddM1	**Dependent on both previous method marks**, uses any clear, valid method to reduce to an equation in a (or less likely, b)
	A1	For correct b
	A1	For correct a
ALT 1	M1	One correct equation as shown oe
	A1	Both correct equations as shown oe
	M1	Clear valid attempt to subtract one equation from the other
	A1	Achieves the two terms shown
	ddM1	**Dependent on both previous method marks,** uses a valid method to eliminate the logs and achieves an equation in b only
	A1	For correct b
	A1	For correct a
ALT 2	M1	Finds a correct equation as shown for the common difference, d
	A1	Correct equation as shown oe
	M1	Correct equation as shown oe (subs to get u_5)
	A1	Correct equation as shown oe
	ddM1	**Dependent on both previous method marks,** eliminates the logs and achieves an equation in a only
	A1	For correct b
	A1	For correct a
ALT 3	M1	One correct equation as shown oe
	A1	Both correct equations as shown oe
	M1	Clear attempt to subtract one equation from the other
	A1	Achieves the correct value for d in any single ln form
	ddM1	**Dependent on both previous method marks,** arrives at a single term log for u_1
	A1	For correct b
	A1	For correct a
	Allow full marks in general for just $b = 4$ and $a = 3$	

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Answer 2 of 3 · Question 3 part whole · 7 marks

2020 NOV · Paper 01 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Series made of logarithms

Answer 3 of 3 · Question 5 part c · 5 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

5 (a)	$[S_\infty =]\dfrac{12}{1-\frac{3}{8}}=\dfrac{96}{5}$	M1 A1 (2)
(b)	$ar^5=12\left(\dfrac{3}{8}\right)^5$	M1
	$=\dfrac{2^2\times3\times3^5}{2^{15}}=\dfrac{3^6}{2^{13}}$ *	M1 A1cso (3)
(c)	e.g. $u_n=12\left(\dfrac{3}{8}\right)^{n-1}$	M1
	$\log_2 u_n=\log_2 12+\log_2\left(\dfrac{3}{8}\right)^{n-1}$	M1
	$\log_2 u_n=\log_2 12+(n-1)\left[\log_2 3-\log_2 8\right]$	M1
	$\log_2 u_n=\log_2 3+2\log_2 2+(n-1)[\log_2 3-3\log_2 2]$	M1
	$\log_2 u_n=n\log_2 3-3n+5$ *	A1 cso (5)
		Total 10 marks

Type MS · Series made of logarithms

Answer 3 of 3 · Question 5 part c · 5 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For the correct use of $\frac{a}{1-r}$
	A1	For the correct value $\frac{96}{5}$
(b)	M1	For the correct use of ar^{n-1}
	M1	For expressing 12 $\left(2^2\times 3\right)$ and 8 $\left(2^3\right)$ as powers of 2
	A1cso	For obtaining the given expression with no errors seen.
(c)	M1	For the correct use of ar^{n-1} and the given values of a and r to give $u_n = 12\left(\frac{3}{8}\right)^{n-1}$
	M1	For taking logs [base 2] of both sides and applying the addition law.
	M1	For applying the power law and subtraction laws to $\log_2\left(\frac{3}{8}\right)^{n-1}$ $\log_2\left(\frac{3}{8}\right)^{n-1} = (n-1)\left[\log_2 3 - \log_2 8\right]$
	M1	For obtaining $\log_2 12 = \log_2 3 + 2\log_2 2$
	A1cso	For obtaining the given equation with no errors seen.
	ALT	
	M1	For the correct use of ar^{n-1} and the given values of a and r to give $u_n = 12\left(\frac{3}{8}\right)^{n-1}$
	M1	For rearranging the equation to obtain: $U_n = 12\left(\frac{3}{8}\right)^{n-1} = 2^2\times 3\times\frac{3^{n-1}}{2^{3(n-1)}} = \frac{3^n}{2^{3n-5}}$
	M1	For taking logs of both sides and applying the addition (subtraction) law. $\log_2 U_n = \log_2 3^n - \log_2\left(3n-5\right)$
	M1	For applying the power law to obtain: $\log_2 U_n = n\log_2 3 - \log_2\left(3n-5\right)$
	A1 cso	For obtaining the given equation with no errors seen.

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Answer 3 of 3 · Question 5 part c · 5 marks

2022 JAN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks